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(%i4) /* Symbols */
declare([C_1, C_2, C_3, C_4], real)$;
declare(h, real)$;
assume(h > 0)$;
depends(D, h);

(%o4) [D(h)]

(%i9) q1: D(h) + C_1 / (h^2) + C_2 / h + C_3 · h + C_4 · (h^2); /* D_true = ... */
q2: subst(h = h / 2, q1);
q4: subst(h = h / 2, q2);
q8: subst(h = h / 2, q4);
q16: subst(h = h / 2, q8);
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q1
$$D(h) + C_4 h^2 + C_3 h + \frac{C_2}{h^2} + \frac{C_1}{h}$$

q2
$$\frac{C_4 h^2}{4} + \frac{C_3 h}{2} + \frac{2 C_2}{h} + \frac{4 C_1}{h^2} + D\left(\frac{h}{2}\right)$$

q4
$$\frac{C_4 h^2}{16} + \frac{C_3 h}{4} + \frac{4 C_2}{h} + \frac{16 C_1}{h^2} + D\left(\frac{h}{4}\right)$$

q8
$$\frac{C_4 h^2}{64} + \frac{C_3 h}{8} + \frac{8 C_2}{h} + \frac{64 C_1}{h^2} + D\left(\frac{h}{8}\right)$$

q16
$$\frac{C_4 h^2}{256} + \frac{C_3 h}{16} + \frac{16 C_2}{h} + \frac{256 C_1}{h^2} + D\left(\frac{h}{16}\right)$$

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(%i10) eq: [q1 = q2, q2 = q4, q4 = q8, q8 = q16]$; /* List of equations */
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(%i11) A: coefmatrix(eq, [C_1, C_2, C_3, C_4]); /* A matrix (Ax = b) */
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A
$$\begin{bmatrix} -\left(\frac{3}{h^2}\right) & -\left(\frac{1}{h}\right) & \frac{h}{2} & \frac{3 h^2}{4} \\ -\left(\frac{12}{h^2}\right) & -\left(\frac{2}{h}\right) & \frac{h}{4} & \frac{3 h^2}{16} \\ -\left(\frac{48}{h^2}\right) & -\left(\frac{4}{h}\right) & \frac{h}{8} & \frac{3 h^2}{64} \\ -\left(\frac{192}{h^2}\right) & -\left(\frac{8}{h}\right) & \frac{h}{16} & \frac{3 h^2}{256} \end{bmatrix}$$

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(%i12) determinant(A); /* It's not zero. A unique solution exists */
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(%o12)
$$\frac{19845}{1024}$$

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(%i13) c: linsolve(eq, [C_1, C_2, C_3, C_4]); /* Cs (not simplified) */
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c
$$[C_1 = -\left(\frac{2 h^2 D(h) + \left(-\left(15 D\left(\frac{h}{2}\right)\right) + 35 D\left(\frac{h}{4}\right) - 30 D\left(\frac{h}{8}\right) + 8 D\left(\frac{h}{16}\right)\right) h^2}{630}\right), C_2 =$$

$$\frac{4 h D(h) + \left(- \left(29 D\left(\frac{h}{2}\right) \right) + 63 D\left(\frac{h}{4}\right) - 46 D\left(\frac{h}{8}\right) + 8 D\left(\frac{h}{16}\right) \right) h}{42}, C_3 =$$

$$\frac{64 D(h) - 368 D\left(\frac{h}{2}\right) + 504 D\left(\frac{h}{4}\right) - 232 D\left(\frac{h}{8}\right) + 32 D\left(\frac{h}{16}\right)}{21 h}, C_4 = -$$

$$\left(\frac{1024 D(h) - 3840 D\left(\frac{h}{2}\right) + 4480 D\left(\frac{h}{4}\right) - 1920 D\left(\frac{h}{8}\right) + 256 D\left(\frac{h}{16}\right)}{315 h^2} \right)$$

(%i14) **abs(factor(subst(c, q1 - D(h))));** /* Absolute Error of D(h): abs(D_true - D(h)) */

(%o14)
$$\frac{\left| D(h) + 54 D\left(\frac{h}{2}\right) - 101 D\left(\frac{h}{4}\right) + 54 D\left(\frac{h}{8}\right) - 8 D\left(\frac{h}{16}\right) \right|}{9}$$